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AI

Machine/deep learning-assisted hemoglobin level prediction using palpebral conjunctival images

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Summary

Palpebral conjunctival hue alteration is used in non-invasive screening for anaemia, whereas it is a qualitative measure. This study constructed machine/deep learning models for predicting haemoglobin values using 150 palpebral conjunctival images taken by a smartphone. The median haemoglobin value was 13.1 g/dL, including 10 patients with <11 g/dL. A segmentation model using U-net was successfully constructed. The segmented images were subjected to non-convolutional neural network (CNN)-based and CNN-based regression models for predicting haemoglobin values. The correlation coefficients between the actual and predicted haemoglobin values were 0.38 and 0.44 in the non-CNN-based and CNN-based models, respectively. The sensitivity and specificity for anaemia detection were 13% and 98% for the non-CNN-based model and 20% and 99% for the CNN-based model. The performance of the CNN-based model did not improve with a mask layer guiding the model's attention towards the conjunctival regions, however, slightly improved with correction by the aspect ratio and exposure time of input images. The gradient-weighted class activation mapping heatmap indicated that the lower half area of the conjunctiva was crucial for haemoglobin value prediction. In conclusion, the CNN-based model had better results than the non-CNN-based model. The prediction accuracy would improve by using more input data with anaemia.

KEYWORDS
artificial intelligence, CNN model, grad-CAM, non-invasive anaemia prediction

INTRODUCTION

Anaemia is caused by various nutritional and haematological diseases, and its prevalence is high worldwide. Anaemia can develop even in children who might appear healthy due to malnutrition, infections, hereditary diseases, and other various diseases.[1] Global anaemia prevalence is ca. 50% in preschool-age children and ca. 25% in school-age children.[2] It can be associated with increased morbidity, such as developmental issues and decreased quality of life and physical function, and mortality.[3] Palpebral conjunctival hue alteration is used as a non-invasive screening indicator for anaemia. Nonetheless, it is generally a qualitative measure that complicates the distinction of borderline cases. Diagnostic precision can vary based on the experience of evaluators without objectivity and reproducibility. Anaemia diagnosis obtained using blood tests is accurate. However, blood examinations have several disadvantages such as their invasive nature, the presence of relatively affordable yet associated medical expenses, and

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the time-consuming nature of the measurements. It also requires visiting a hospital.

The application of artificial intelligence in conjunctival image analysis for obtaining a non-invasive, cost-effective and rapid anaemia diagnosis can be a promising clinical tool in various healthcare environments. If such a system is practical, it can facilitate the measurement of haemoglobin values at home, thereby addressing the above-mentioned potential disadvantages associated with anaemia. Indeed, previous studies constructed machine/deep learning models for non-invasive anaemia diagnosis (Table S1).[4–14] These studies utilised various segmentation and prediction algorithms, including machine/deep learning, and achieved 57%–100% of sensitivity and 71%–100% of specificity. For example, Suner et al. used semi-automatic segmentation of the conjunctival region in images captured with smartphones.[9] They developed a regression model based on machine learning for factors such as the colour tone of the conjunctival regions and achieved successful anaemia diagnosis with a sensitivity of 73% and a specificity of 83%. Further, Ghosal et al. constructed a regression model that identifies haemoglobin values based on the colour tone of the conjunctiva on images taken using smartphones.[10] Their model had an accuracy of ±0.32 g/dL for estimating haemoglobin values ($R^2 = 0.95$) and a sensitivity of 89% for anaemia detection. Dimauro et al. developed a dedicated device for capturing conjunctival images.[7] Using the images taken with this device, a model was created to predict haemoglobin values and achieve favourable outcomes ($R^2 = 0.65$). Appiahene et al. constructed a segmentation model for the conjunctival region using convolutional neural networks (CNN) and developed a regression model based on machine learning for conjunctival colour tone.[12] These sequential models were embedded into a smartphone application, achieving a prediction accuracy of 92.5% for the presence of anaemia. Asare et al. compared the usefulness of CNN and conventional machine learning methods such as Naïve Bayes for predicting anaemia using images of the conjunctiva, fingernails and palms, showing that CNN had the highest accuracy.[14]

Thus, while various models were utilised for non-invasive anaemia prediction, studies on anaemia prediction using palpebral conjunctival images with deep learning are limited. Moreover, which parts or features of the palpebral conjunctival images in deep learning models are important for predicting anaemia are not yet identified. The current study aimed to develop deep learning models for segmenting conjunctival images captured with a smartphone. The segmented images were subjected to both non-CNN-based and CNN-based regression models for predicting haemoglobin levels. Further, using gradient-weighted class activation mapping (Grad-CAM), the vital regions and features in the analysis of conjunctival images that are essential for anaemia prediction in the CNN-based model were explored.

MATERIALS AND METHODS

Patient recruitment and data collection

This prospective observational study was a collaborative research project between the University of Tokyo and LPIXEL Inc. (Tokyo, Japan). The Research Ethics Committee of the University of Tokyo approved this study (approval number: 2022232NI). Patients who attended outpatient appointments or were admitted to the paediatric department in the University of Tokyo Hospital between 30 November 2022 and 31 August 2023, and those who underwent blood tests including haemoglobin values for clinical purposes were eligible for this research. A written informed consent was obtained from the patients or their legal guardians.

The patients or their guardians retracted the lower eyelid, and a research collaborator captured one image of each eye. To prevent patient identification, the field of each image was limited to one eye and its minimally extensive surrounding area. Images were obtained under indoor lighting without built-in flash in JPEG format using the standard camera application on iPhone 14 (Apple). Data on the results of blood tests carried out on the same day as the images were obtained were collected from the patients' records.

To develop the segmentation model, annotations of the palpebral conjunctival area were performed on images from the first 100 of 167 patients using QuPath.[15] Patients with ophthalmic conditions which might alter the colour of the palpebral conjunctiva were excluded.

Conjunctiva segmentation model

Residual U-Net[16] was implemented by modifying U-Net[17] from the segmentation_models_pytorch library (github.com/qubvel/segmentation_models.pytorch; Figure 1). The architecture utilised an EfficientNet-B0 backbone[18] with three input channels. A composite loss function, combining Binary Cross-Entropy, Dice loss and Focal loss[19] at a ratio of 1:2:1 was used.

Our segmentation model was trained using 180 images from 90 patients. Then, a fivefold cross-validation was conducted, assigning 36 images to each fold. The left and right eye images from the same patients were placed within the same fold. Augmentation techniques, including random cropping, horizontal flipping, affine transforms, random changes in brightness and contrast, gamma correction, the addition of Gaussian noise, and the application of Gaussian or motion blur to the images by the Albumentations library, were applied during training cycles.[20] Augmented images were normalised and used as input for the model. The backbone of the model was pretrained on ImageNet.[21] The RAdam optimiser[22] with an initial learning rate of 10^{-4} scheduled by Cosine Annealing[23] implemented as CosLRScheduler in the timm library (github.com/rwightman/pytorch-image-models) was utilised. The training

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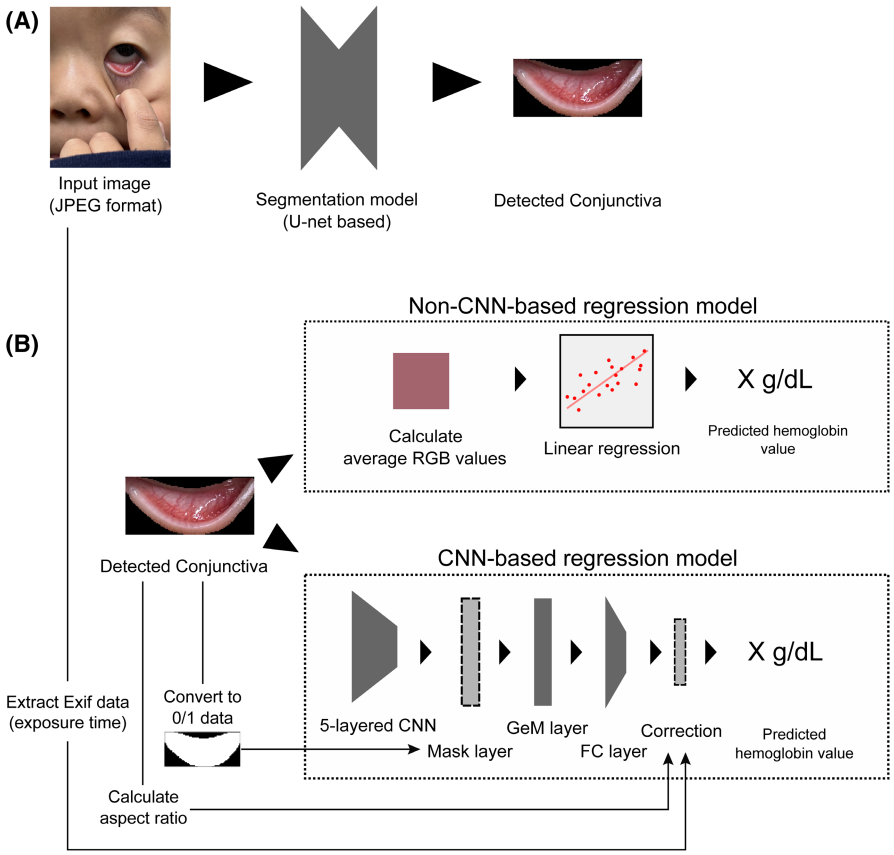


FIGURE 1 The model structure in our study. The conjunctival region was detected by the segmentation model from the image taken using a smartphone camera (A). The detected conjunctiva was used as an input for both the non-CNN- and CNN-based regression models (B). CNN, convolutional neural network; Exif, exchangeable image file format; FC layer, fully-connected layer; GeM layer, generalised mean pooling layer.

was conducted for 50 epochs with a batch size of 10. The model with the highest intersection over union (IoU) on the validation dataset of each fold was adopted. If multiple regions were detected, the largest one was considered as the conjunctiva. The conjunctivae on 300 images from 150 patients, including 180 images used for the model training and 120 additional images not used for the training, were detected by the segmentation model and used as input for the subsequent models.

Haemoglobin value prediction using the multiple linear regression model

A multiple linear regression model using the least squares method was created to infer haemoglobin values from the R, G and B values of the conjunctiva. In total, 300 conjunctivae from 150 patients detected using the segmentation model were utilised to obtain the colour of the conjunctiva. From each patient, the average R, G and B values of the left and right palpebral conjunctiva were obtained and used as the explanatory variables of the model. The actual haemoglobin value of the patient was used as the response variable. The 'LinearRegression' method implemented in scikit learn[24] was utilised for multiple linear regression, calculating both coefficient and intercept simultaneously. The regression analysis was performed via a fivefold cross-validation to evaluate the model, assigning 60 images to each fold. The left and right eye images from the same patient were placed in the same fold.

Haemoglobin value prediction using the CNN-based model

A CNN regression model was developed to assess haemoglobin values based on the detection results of the conjunctiva segmentation model. A five-layered CNN was implemented, followed by a Generalised Mean (GeM) pooling layer,[25] where the output of the layer was flattened and connected to a fully-connected layer for predicting haemoglobin values (Figure 1). Mean Square Error (MSE) loss was utilised for the loss function. For the alternative model, a mask layer was introduced between the output of CNN and the input of the GeM layer to guide the model's attention exclusively towards the conjunctiva (Figure 1). As a correction for the output value, the exposure time when the image was taken and the aspect ratio of the segmented conjunctiva were considered. Although all images were taken in rooms with the same type of lighting, considering that the patients' position relative to the lighting and the direction of the face could affect the colour tones

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in the images, the exposure time was used for correction. The exposure time was obtained from the Exchangeable Image File Format (Exif) data of the original JPEG image. The aspect ratio was calculated from the predicted conjunctiva by dividing the height by the width, where a low aspect ratio indicates insufficient retraction of the eyelid. The predicted haemoglobin value was concatenated with either or both values and subsequently connected to a fully-connected layer to obtain the corrected haemoglobin values (Figure 1).

Each regression model was trained using 300 conjunctival images from 150 patients, which were obtained by our segmentation model but dilated to include the surrounding area of the conjunctiva using a 3×3 kernel. Then, a fivefold cross-validation was conducted, assigning 60 images to each fold. The left and right eye images from the same patient were placed in the same fold. Augmentation techniques, including horizontal and vertical flipping, affine transforms, elastic transform or grid distortion, coarse dropout, and the application of Gaussian or motion blur to the images by the Albumentations library, were used during training cycle.[20] Augmented images were normalised and used as input for the model. The AdamW optimiser[26] with an initial learning rate of 10^{-3} scheduled by Cosine Annealing[23] implemented as CosLRScheduler in the timm library was used. The training was conducted for 60 epochs with a batch size of 32. The model with the best MSE performance on the validation dataset of each fold was adopted. To evaluate the model, haemoglobin value prediction was performed by obtaining the average of the separately predicted haemoglobin values from each left and right eye image of the patients, and compared with the actual haemoglobin value of the patient.

Grad-CAM[27] was used to visualise the regions where our regression models were utilised for predictions. Grad-CAM suggests the areas of an input image that contribute to the model's decision by using gradients in the final convolutional layer of the CNN. The results were shown as heatmaps.

RESULTS

Characteristics of the patients

In total, 167 patients participated in this study. Among them, 17 with poor-quality conjunctival images were excluded from the research. Finally, 150 patients were included in the current analysis. Table 1 shows the characteristics of the eligible patients. Among the patients, 83 (55.3%) were male and the median age of the participants was 11 years. Haematological/oncological diseases ($n = 145$, 96.7%) were the most prevalent underlying conditions, followed by non-cyanotic heart diseases ($n = 2$, 1.3%), cyanotic heart diseases ($n = 1$, 0.7%) and other conditions ($n = 2$, 1.3%). The median haemoglobin value of all patients was 13.1 (7.7–16.8) g/dL, and the haemoglobin values of 10 (6.7%) patients were <11 g/dL (Figure 2).

TABLE 1 Characteristics of the eligible patients ($N = 150$).

Characteristics	*N* (%) or median (range)
Age (years)	11 (1–37)
Sex	
Male	83 (55.3)
Female	67 (44.7)
Underlying conditions	
Haematological/oncological disease	145 (96.7)
Non-cyanotic cardiac disease	2 (1.3)
Cyanotic cardiac disease	1 (0.7)
Endocrine disease	1 (0.7)
Premature born	1 (0.7)
Haemoglobin value (g/dL)	13.1 (7.7–16.8)

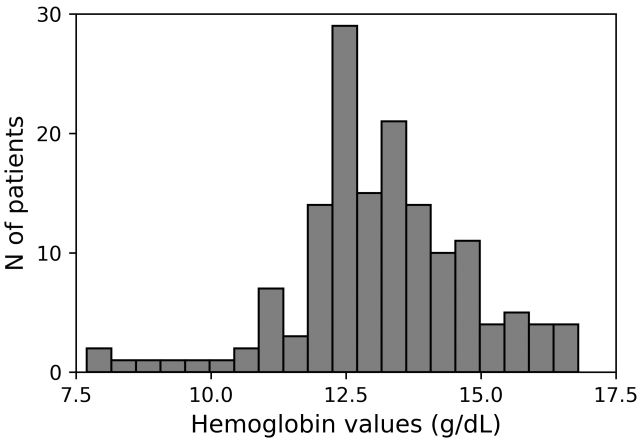


FIGURE 2 The distribution of haemoglobin values ($N = 150$).

Segmentation

The IoU of 90 patients with annotated conjunctiva regions was assessed using fivefold cross-validation. The average IoU of all folds was 0.84 (0.83–0.86). In one image, a pinkish face mask was detected, and in another, a lip was detected (Figure 3). These two images corresponded to the images with a lower IoU, with the first having the lowest and the second having the third lowest. Further, images with a slender conjunctiva, which indicated that the eyelid was not adequately retracted, or those obscured by eyelashes were more likely to have a low IoU.

Haemoglobin value prediction using the multiple linear regression model

The segmentation model was extended to 300 images from 150 individuals for haemoglobin value prediction. The

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average R, G and B values of each image were utilised as explanatory variables to predict haemoglobin values. The performance of the linear regression model was assessed via a fivefold cross-validation. The correlation coefficient of the training and testing dataset ranged from 0.42 to 0.56 and from 0 to 0.58, respectively (Table 2). The correlation coefficient between the predicted haemoglobin values of the testing dataset, obtained from each fold's regression equation, and the actual haemoglobin values was 0.38 ($R^2 = 0.14$, Figure 4). The Bland–Altman plot analysis indicated that the predicted values tend to be underestimated at higher measurements and overestimated at lower measurements (Figure S1). The sensitivity and specificity of the model with a cut-off of the haemoglobin value of 11 g/dL were 13% and 98%, respectively. The true positive and false positive rates with increasing predicted haemoglobin values are depicted in a receiver operator characteristic (ROC) curve, for predicting actual haemoglobin value of <11 g/dL (Figure S2). The area under the curve (AUC) was 0.82 for the non-CNN-based model.

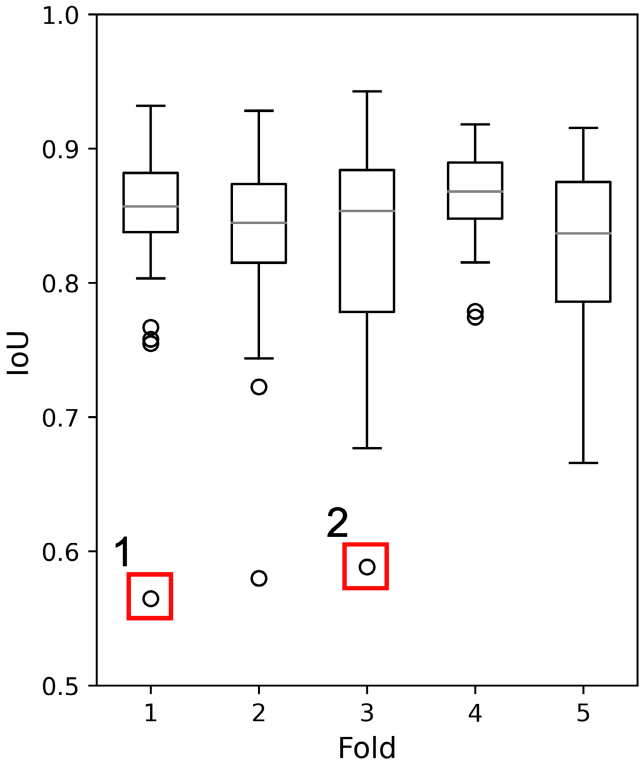


FIGURE 3 Comparison of the intersection over union (IoU) for each fold of the segmentation model. The red boxes in the graph indicate the images where the IoU was low due to false detections of the non-conjunctival regions. 1: An image with a pinkish face mask detected as the conjunctiva. 2: An image with a lip detected as the conjunctiva.

Haemoglobin value prediction using the CNN-based regression model

The subsequent analysis with the CNN-based regression model also covered the 150 patients. The performance was assessed via a fivefold cross-validation. For the model without the mask layer, the correlation coefficient for the overall predicted data was 0.44, with a tendency for predictions to be less successful if the actual haemoglobin values were <11 or >15 g/dL (Figure 5). The sensitivity and specificity of the model using a haemoglobin value of 11 g/dL as the cut-off were 20% and 99%, respectively. The correlation coefficient for each fold ranged from 0.26 to 0.63 (Table 2). For the model with the mask layer, the correlation coefficient for the overall predicted data was 0.39, with a tendency for predictions to be less successful if the actual haemoglobin values were <11 or >15 g/dL (Figure 5). The sensitivity and specificity of the model using a haemoglobin value of 11 g/dL as

TABLE 2 Correlation coefficient for each fold of the models.

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	All
Non-CNN-based model						
Linear regression model	0.57	0.54	0.49	0.20	0.17	0.38
CNN-based models						
Without the mask layer						
No correction	0.61	0.46	0.63	0.31	0.26	0.44
Aspect ratio correction	0.47	0.52	0.60	0.10	−0.01	0.35
Exposure time correction	0.47	0.58	0.54	0.26	0.37	0.39
Both corrections	0.66	0.55	0.61	0.10	0.39	0.45
With the mask layer						
No correction	0.75	0.59	0.62	−0.04	0.04	0.39
Aspect ratio correction	0.60	0.34	0.54	0.35	0.25	0.39
Exposure time correction	0.65	0.26	0.61	−0.01	0.28	0.36
Both corrections	0.61	0.41	0.57	0.14	0.29	0.40

Note: All indicates the correlation coefficient when the results of each fold are plotted together.
Abbreviation: CNN, convolutional neural network.

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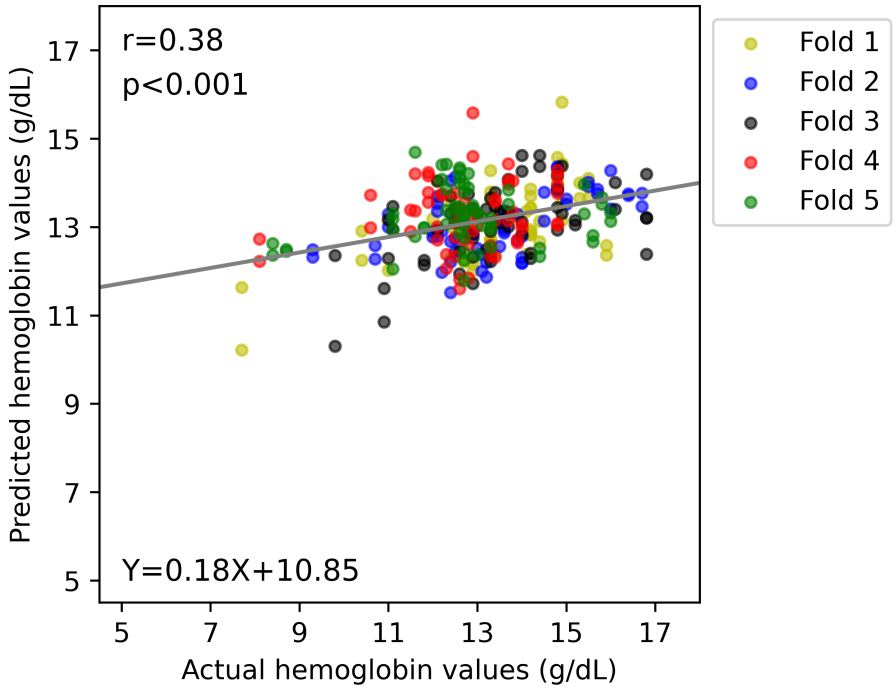


FIGURE 4 The predicted haemoglobin values estimated using the non-CNN-based linear regression model.

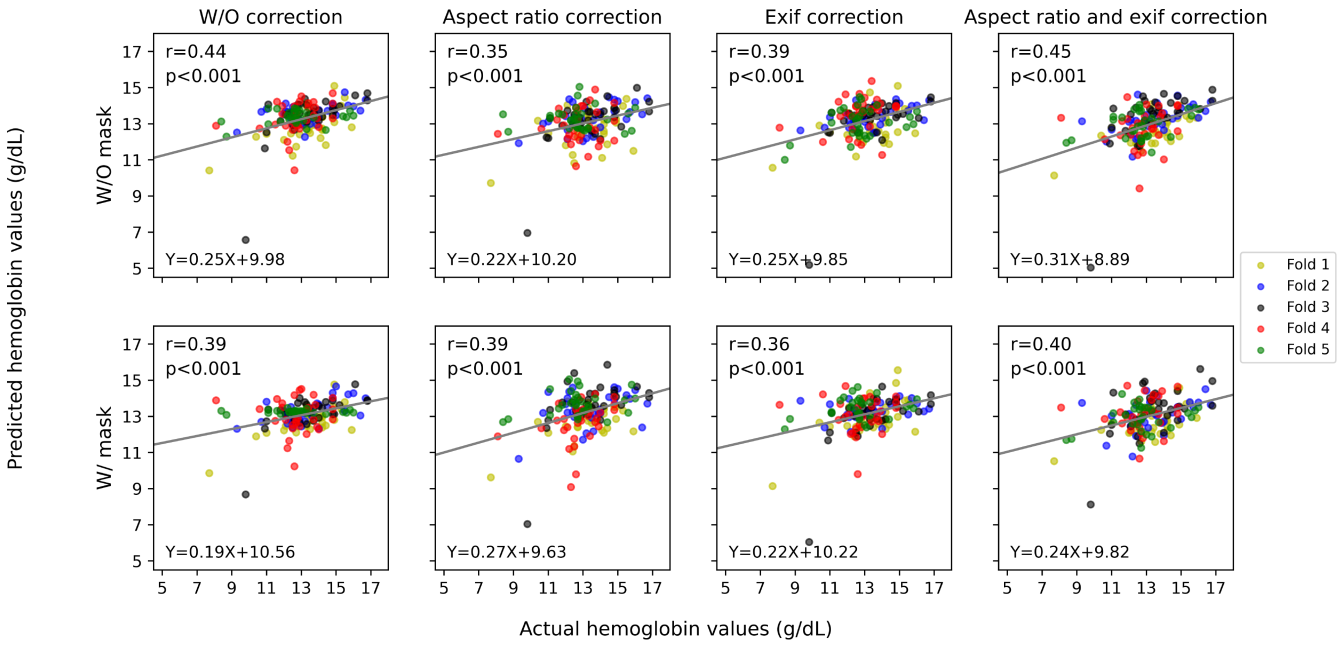


FIGURE 5 The predicted haemoglobin values (g/dL) estimated using the CNN-based regression models. The top and bottom rows show the results of the models with and without the mask layer, respectively. The first column displays the results of the model without any corrections. Meanwhile, the subsequent columns show the results of the model with corrections: correction by aspect ratio of the conjunctiva, correction by exposure time obtained from exchangeable image file format data (Exif), and both corrections.

the cut-off were 20% and 99%, respectively. The correlation coefficient for each fold ranged from −0.04 to 0.75 (Table 2).

Considering the potential impact of excessive eyelid retraction and lighting on the conjunctival colour, the aspect ratio of the conjunctival images and exposure time during image acquisition were included as correction factors in the learning process. The correlation coefficient for prediction using the model without the mask layer corrected by the aspect ratio was 0.35. Meanwhile, the correlation coefficient corrected by the exposure time

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was 0.39 (Table 2). The correlation coefficient of correction by both the aspect ratio and exposure time was 0.45 (Table 2). For the model with the mask layer, the correlation coefficients from each model corrected by the aspect ratio, exposure time and both factors were 0.39, 0.36 and 0.40, respectively (Table 2). The model without the mask layer corrected by both the aspect ratio and exposure time achieved the highest correlation coefficient (Table 2). The Bland–Altman plot analysis for this model showed that the predicted values tend to be underestimated in higher values and overestimated in lower values although all samples except for one were within the 95% limits of agreement (Figure S1). The sensitivity and specificity with a cut-off of the haemoglobin value of 11 g/dL were 20% and 98%, respectively, and the AUC of the ROC curve was 0.74 (Figure S2).

Overall, the mask layer did not have a significant effect on the model based on its overall correlation coefficient. However, the fold with the highest correlation coefficient was that from the model with the mask layer. Further, correction by both the aspect ratio and exposure time had no significant effect on the model. Only the correlation coefficient for haemoglobin value predictions slightly increased. However, the specificity slightly decreased compared with that of the model without correction. The CNN-based model had better correlation coefficients and sensitivity/specificity than the non-CNN-based linear regression model.

CNN model visualisation using grad-CAM

In the two models, with or without the mask layer, for the folds with a high accuracy in estimating haemoglobin values, the Grad-CAM heatmap had a significant intensity in the lower half to the marginal areas of the conjunctiva. Hence, these regions play an important role in measuring haemoglobin values (Figure 6A). Conversely, in the folds with a lower estimation accuracy, attention was directed to not only the conjunctiva region but also areas outside of the conjunctiva, or there was no focus on the conjunctiva region at all, with attention solely on the external areas (Figure 6B,C). The mask layer was implemented to guide the model's attention exclusively towards the conjunctiva; nevertheless, it did not have a significant impact for the purpose.

DISCUSSION

Although there have been studies on non-invasive anaemia diagnosis using AI, most of them have been conducted in Africa and South Asia,[9,10,12,14] and this is the first research performed in East Asia. The IoU of the segmentation was 0.84, which was comparable to that of previous studies. The current linear regression model was inferior to that in a previous study.[10] Furthermore, CNN-based models, which are expected to have higher performance than non-CNN-based model by being a more complex model, also exhibited a trend

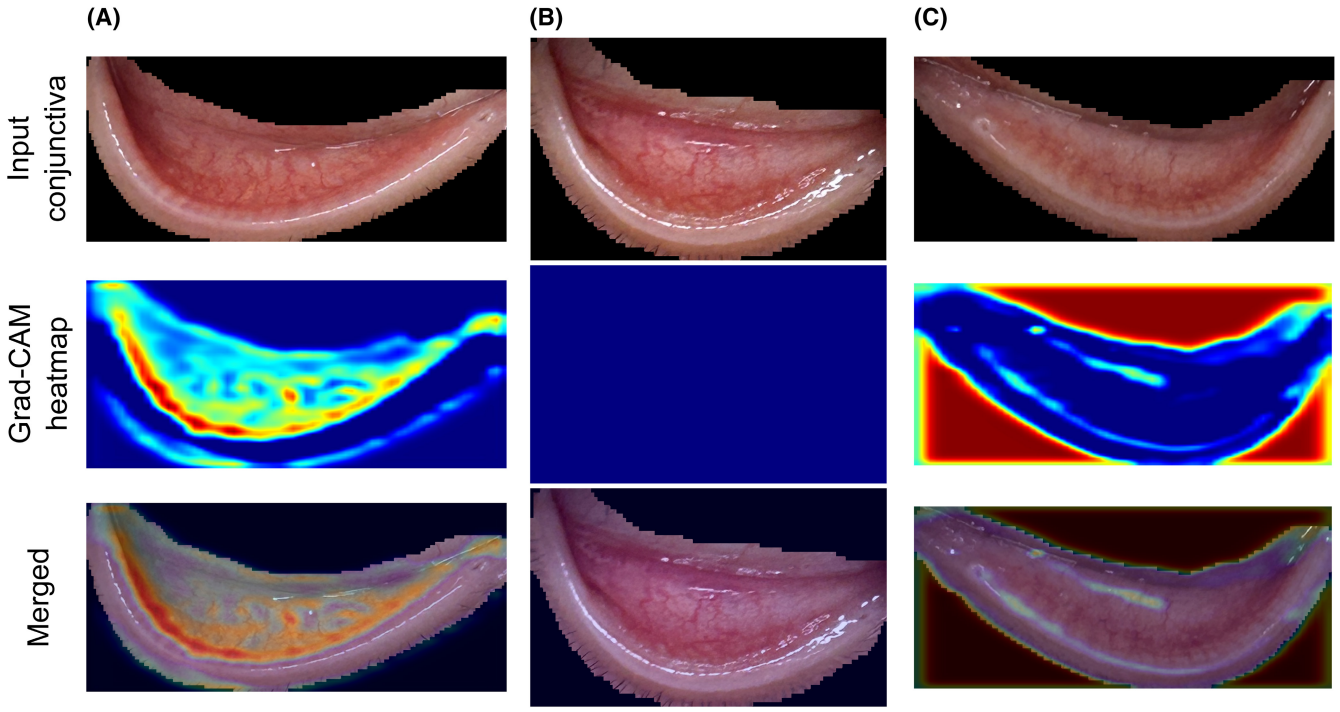


FIGURE 6 The visualisation results of the CNN-based regression model using gradient-weighted class activation mapping (Grad-CAM). The top and middle rows show the input conjunctiva images and the heatmaps obtained from Grad-CAM, respectively. The bottom row shows the merged images of the conjunctiva and the heatmap. The regions in red contribute more significantly to the model's predictions, whereas regions in blue contribute less. Examples showing attention at the ideal position (A), no attention anywhere (B) and attention at unnecessary locations (C).

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of lower correlation than the previous studies, although with a higher correlation coefficient than that of our non-CNN-based model.

The suboptimal precision in anaemia prediction may be attributed to the following reasons. First, the number of patients ($n = 150$) used in model construction was limited. Second, only 6.7% of participants had a haemoglobin value of <11 g/dL. Previous research has shown that the CNN model outperformed conventional machine learning models in predicting anaemia.[14] Therefore, the accuracy of our CNN-based model might improve by including a higher number of patients with low haemoglobin values. Image quality is another factor that may affect the performance of the model. In the current study, some images included noses, face masks, or other components, resulting in less consistent conditions during image capture compared with those in previous studies. In addition, since the patients were mostly children, their conjunctiva might not have been exposed properly. Dimauro et al. developed their model by creating a dedicated device to standardise the capture conditions.[7] Further, Ghosal et al. captured only the area around the eye and excluded improperly captured images such as those with insufficiently exposed conjunctival region.[10] Image processing by JPEG compression also might be a potential factor for the low accuracy because JPEG compression produces various artefacts.[28,29] Thus, utilizing uncompressed images with a focus on the conjunctiva and its surrounding areas only and standardizing the capture conditions might be required for developing a model with good performance. On the other hand, considering the development of a user-friendly interface that is not restricted by equipment or location, we believe that it is worthwhile to further advance our methodology.

One novel aspect of this study is that it showed the regions of the palpebral conjunctival images that should be the focus of the CNN-based model. In the folds where haemoglobin value prediction was successful, the model focused on the lower half to the marginal areas of the palpebral conjunctiva region. This observation was concordant with results from a previous study, where haemoglobin levels were inferred from the degree of redness both in the upper and lower half of the conjunctival regions.[6] The lower half of the palpebral conjunctiva region can be significant because the lower half is likely to be less affected by colour changes attributed to retraction even though the upper half region closer to the eyeball is at risk of colour changes based on the degree of eyelid retraction. However, in folds with a poor accuracy, the focus was mistakenly placed solely on areas outside the palpebral conjunctiva region. Constructing an adequate model may be complex due to biases in data quality across the folds, which can be affected by factors such as the degree of eyelid retraction, the overlapping of eyelashes on the palpebral conjunctiva, and reflections from lighting. This issue was addressed by installing a mask layer to help our model focus exclusively on the conjunctiva, although it was ineffective. These issues might be resolved by increasing the number of patients with anaemia.

In conclusion, the CNN-based model yielded better results than the non-CNN-based model. Therefore, utilizing fine-tuned pretrained CNN models can be more suitable for creating models with few samples or low-quality images. Considering the limited number of research on anaemia prediction models using deep learning, there is potential to create a deep learning model with better precision by increasing the number of samples.

AUTHOR CONTRIBUTIONS

Conceptualisation: S. Kato, M. Kato; Financial support: M. Kato; Collection of materials and data: S. Kato, M. Hidaka, S. Inoue, M. Sekiguchi, N. Adachi, K. Sato, M. Kato; Data curation and analysis: S. Kato, K. Chagi, Y. Takagi, H. Kawai; Manuscript writing: S. Kato, K. Chagi; Manuscript editing: Y. Takagi, H. Kawai, M. Hidaka, S. Inoue, M. Sekiguchi, N. Adachi, K. Sato, M. Kato; Final approval of manuscript: All authors; Accountable for all aspects of the work: All authors.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

For original data, please contact katom-tky@umin.ac.jp.

ETHICS APPROVAL STATEMENT

The Research Ethics Committee of the University of Tokyo approved this study (approval number: 2022232NI).

PATIENT CONSENT STATEMENT

A written informed consent was obtained from the patients or their legal guardians.

PERMISSION TO REPRODUCE MATERIAL FROM OTHER SOURCES

Not applicable.

CLINICAL TRIAL REGISTRATION

Not applicable.

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Additional supporting information can be found online in the Supporting Information section at the end of this article.

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